

Chacha Chen

CONTACT +1 (814)-699-2243 Computer Science Department
 chacha@uchicago.edu The John Crerar Library, 5730 S Ellis Ave
 <https://chacha-chen.github.io> Chicago, IL 60637

I've always been driven to build better AI that works well with/for people. My research spans natural language processing, computational social sciences, and data mining. Recently, I've been focusing on multimodal LLMs and their applications in the healthcare domain.

EDUCATION **University of Chicago** 2021 – present
 Ph.D. in Computer Science
 Advisor: [Chenhao Tan](#)

Pennsylvania State University 2019 – 2020
Master in Informatics
Advisors: [Ting-Hao \(Kenneth\) Huang](#), [Zhenhui \(Jessie\) Li](#)

Shanghai Jiao Tong University 2015 – 2019
Bachelor of Science in Computer Science

PROFESSIONAL EXPERIENCE **Microsoft Research**, Research Intern Jun. 2023 –
 Mentors: [Siddharth Suri](#), [Scott Counts](#), [Longqi Yang](#).

Amazon AWS, Applied Scientist Intern Jun. 2022 – Aug. 2022
Mentors: Sandesh Swamy, Narges Tabari, and [Rashmi Gangadharaiah](#)

IQVIA, Machine Learning Research Intern Jun. 2020 – Aug. 2020
Mentors: [Cao \(Danica\) Xiao](#), [Fenglong Ma](#) (PSU), and [Jimeng Sun](#) (UIUC).

PUBLICATIONS  [Google scholar](#)

Preprints

1. Siddharth Suri, Scott Counts, Leijie Wang, **Chacha Chen**, Mengting Wan, Tara Safavi, Jennifer Neville, Chirag Shah, Ryen W White, Reid Andersen, Georg Buscher, Sathish Manivannan, Nagu Rangan, Longqi Yang
The Use of Generative Search Engines for Knowledge Work and Complex Tasks

Conference Papers

- P.1 Yuyang Jiang*, **Chacha Chen***, Dang Nguyen, Benjamin M. Mervak, Chenhao Tan. GPT-4V Cannot Generate Radiology Reports Yet. In Machine Learning for Health (**Findings of ML4H 2024**).
- P.2 Nguyen Dang, **Chacha Chen**, He He, and Chenhao Tan. Pragmatic Radiology Report Generation. In Machine Learning for Health (**ML4H 2023**).
- P.3 **Chacha Chen***, Shi Feng*, Amit Sharma, Chenhao Tan. Machine Explanations and Human Understanding. Accepted at Transactions on Machine Learning Research (**TMLR 2023**); presented at **FAccT 2023**; Best paper award at the ICML 2022 Workshop on Human-Machine Collaboration and Teaming.

- P.4 Vivian Lai, **Chacha Chen**, Alison Smith-Renner, Vera Liao, and Chenhao Tan. Towards a Science of Human-AI Decision Making: A Survey of Empirical Studies. (**FAccT 2023**).
- P.5 Vivian Lai*, Yiming Zhang*, **Chacha Chen**, Vera Liao, and Chenhao Tan. Selective Explanations: Leveraging Human Input to Align Explainable AI. In Proceedings of the 26th ACM Conference on Computer-Supported Cooperative Work and Social Computing (**CSCW 2023**).
- P.6 Han Liu, Yizhou Tian, **Chacha Chen**, Shi Feng, Yuxin Chen, Chenhao Tan. Learning Human-Compatible Representations for Case-Based Decision Support. In Proceedings of the 11th International Conference on Learning Representations (**ICLR 2023**).
- P.7 Sandesh Swamy, Narges Tabari, **Chacha Chen**, Rashmi Gangadharaiyah. Contextual Dynamic Prompting for Response Generation in Task-oriented Dialog Systems. In Proceedings of the 17th Conference of the European Chapter of the Association for Computational Linguistics (**EACL 2023**).
- P.8 Chi-Yang Hsu, Yun-Wei Chu, Vincent Chen, Kuan-Chieh Lo, **Chacha Chen**, Ting-Hao Huang, Lun-Wei Ku. Learning to Rank Visual Stories From Human Ranking Data. In Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (**ACL 2022**).
- P.9 Guanjie Zheng, Chang Liu, Hua Wei, Porter Jenkins, **Chacha Chen**, Tao Wen, Zhenhui Li. Knowledge-based Residual Learning. In Proceedings of the The 30th International Joint Conference on Artificial Intelligence (**IJCAI 2021**).
- P.10 **Chacha Chen**, Junjie Liang, Fenglong Ma, Lucas Glass, Jimeng Sun, Cao Xiao, UNITE: Uncertainty-based Health Risk Prediction Leveraging Multi-sourced Data, In Proceedings of The Web Conference 2021 (**WWW 2021**).
- P.11 Guanjie Zheng, Chang Liu, Hua Wei, **Chacha Chen**, Zhenhui Li, Rebuilding City-Wide Traffic from Multi-Source Data. In Proceedings of 37th IEEE International Conference on Data Engineering (**ICDE 2021**).
- P.12 Hua Wei, **Chacha Chen**, Chang Liu, Guanjie Zheng and Zhenhui Li, ImIn-GAIL: Learning to Simulate with Imitation-Interpolation on Sparse Trajectory Data. In Proceedings of European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases (**ECML-PKDD 2020**).
 *Best Applied Data Science Paper Award*
- P.13 **Chacha Chen**, Hua Wei, Nan Xu, Guanjie Zheng, Ming Yang, Yuanhao Xiong, Kai Xu and Zhenhui Li, Toward A Thousand Lights: Decentralized Deep Reinforcement Learning for Large-Scale Traffic Signal Control. In Proceedings of the Thirty-Fourth AAAI Conference on Artificial Intelligence (**AAAI 2020**).
- P.14 Hua Wei, Nan Xu, Huichu Zhang, Guanjie Zheng, Xinshi Zang, **Chacha Chen**, Weinan Zhang, Yanmin Zhu, Kai Xu and Zhenhui Li, CoLight: Learning Network-level Cooperation for Traffic Signal Control. In Proceedings of the 2019 ACM on Conference on Information and Knowledge Management (**CIKM 2019**).
- P.15 Hua Wei, **Chacha Chen**, Guanjie Zheng, Kan Wu, Vikash V. Gayah, Kai Xu and Zhenhui Li, PressLight: Learning Max Pressure Control to Coordinate Traffic Signals in Arterial Network. In Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (**KDD 2019**).

Workshop Papers

- W.1 **Chacha Chen***, Shi Feng*, Amit Sharma, Chenhao Tan, Machine Explanations and Human Understanding, ICML 2022 Workshop on Human-Machine Collaboration and Teaming (**HMCaT, ICML 2022**).
 *Best Paper Award*

W.2 Tongan Cai*, **Chacha Chen***, Ting-Hao K. Huang, Frank E. Ritter, What Makes A Good Reference Manager? A Quantitative Analysis of Bibliography Management Applications, Asian CHI Symposium 2021 (**AsianCHI2021**).

 **Best Paper Award**

W.3 **Chacha Chen**, Guanjie Zheng, Hua Wei, Zhenhui Li, Physics-informed Generative Adversarial Networks. In Proceedings of the NeurIPS Workshop on Interpretable Inductive Biases and Physically Structured Learning (**IIBPSL, NeurIPS 2020**).

AWARDS	Crerar Fellowship, <i>The University of Chicago</i>	Feb. 2021
	AAAI-20 VISA Research Scholarship	Feb. 2020
	AAAI-20 Student Travel Grant	Feb. 2020
	Top 25 in the finalist, <i>KDD Cup 2020 LDR Competition</i>	Aug. 2020
	Distinguished Bachelor Thesis, <i>Shanghai Jiao Tong University</i>	Jun. 2019
	Academic Excellence Scholarship, <i>Shanghai Jiao Tong University</i>	Sep. 2016
COMMUNITY ACTIVITIES & SERVICE	Organizing committee of FAccT conference 2024.	
	Organizer of UChicago Computer Science Student Seminar 2021-2022.	
	Student Representative of Computer Science department at Dean's Student Advisory Committee 2020-2021.	
TALKS	<i>Machine Explanations and Human Understanding</i>	
	Human-Centric AI group at NEC Labs Europe in Heidelberg, Germany, 2022.	
	<i>Interpretable Machine Learning</i>	
	Data Mining and Machine Learning seminar at the Pennsylvania State University, 2020	
TEACHING EXPERIENCE	The University of Chicago	
	Teaching Assistant	Spring 2024
	CMSC 25300: Mathematical Foundations of Machine Learning	
	Prof. Haifeng Xia, Prof. Hannah Morgan	
	Pennsylvania State University	
	Teaching Assistant	Fall 2020
	IST 261 (Java Course): Application Development Design Studio I	
	Prof. Kenneth Huang	